



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)

Final Exam, Trimester: Spring 2026

Course Code: CSE 3521 Course Title: Database Management Systems

Total Marks: 40 Duration: 2 hours

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

- 1 a) Consider the relation $R(A, B, C, D, E, F, G)$ with functional dependencies: **3+2+3 = 8**

$A \rightarrow DE$

$CD \rightarrow A$

$BF \rightarrow CG$

$AEG \rightarrow C$

$EF \rightarrow BC$

$F \rightarrow G$

i) **Find** out all possible candidate keys.

ii) **Check** and **justify** in which normal form the relation is.

iii) **Compute** the Minimal Cover of the Relation, R .

- b) A relation $R(A, B, C, D, E)$ with functional dependencies: **3x2=6**

$A \rightarrow B$

$AC \rightarrow DE$

$E \rightarrow D$

$B \rightarrow AC$

is decomposed into $R_1(A, B, C)$, $R_2(D, E)$, and $R_3(A, C, E)$.

i) **Find** whether the decomposition is lossless or not.

ii) Based on the decomposition, is it dependency-preserving or not?

iii) **Find** whether $A \rightarrow CDE$ is valid or not using Armstrong's axioms. [mention the axiom applied at each step].

- 2 a) A hospital stores its patient health records in a system sorted by Patient ID. Due to a sudden shortage of storage resources, minimizing storage space is the top priority. The hospital is willing to accept slightly longer search times as a trade-off for reduced storage usage.

Which type of indexing (Sparse or Dense) should the team implement? Justify your answer based on the given constraints. Also, provide an example of how searching should be performed using your recommended indexing method.

- b) **Construct** a B+ Tree of order 4 using the following data inserted sequentially and show all necessary steps while inserting. 6

M, D, S, H, F, B, K, P, W, G, E, J

- 3 a) How does hashing reduce search time in databases? Is it always suitable for searching for records? Justify your answer with appropriate examples. 2
b) Consider the following search keys: 8

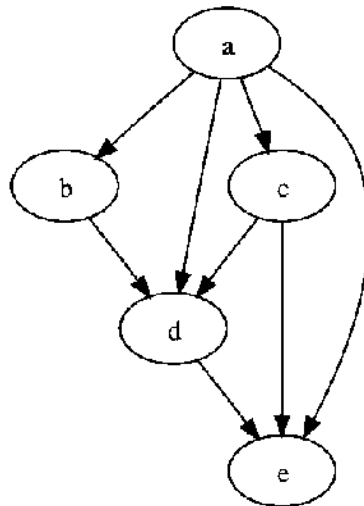
60, 36, 51, 33, 54, 48, 44, 32, 37, 39

Use the hash function:

$$h(x) = x \% 32$$

Insert the hash values one by one into an extensible hashing structure using LSB mapping. Assume the bucket capacity is 3, and the initial local depth and global depth are both 1. Show the final directory, bucket contents, global depth, and local depth of each bucket step by step.

- 4 a) The following is a precedence graph for a schedule Z of transactions.



Can this schedule Z be serializable? Justify your answer. If it is possible, **find** the serializability order; otherwise, explain why it is not. 4

- b) **Explain** the dirty read problem with an example. 2